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Covert Timing Channels based on HTTP Cache Headers (Special Edition f...

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Denis Kolegov

, PhD, associate professor at Tomsk State University

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- 1. COVERT TIMING CHANNELS BASED ON HTTP CACHE HEADERS Denis Kolegov, Oleg Broslavsky, Nikita Oleksov F5 Networks Tomsk State University Information Security and Cryptography Department Top 10 Web Hacking Techniques of 2014 Special Edition ZeroNights (13-14 November 2014) Moscow, Russia SibeCrypt (8-13 September 2014) Ekaterinburg, Russia
- 2. Who we are? • Denis Kolegov – Sr. security test engineer at F5 Networks – PhD, associate professor at Tomsk State University Information Security and Cryptography Department • Oleg Broslavsky – 3rd year student at Tomsk State University Information Security and Cryptography Department – Member of TSU's SiBears Capture the Flag team • Nikita Oleksov – 3rd year

student at Tomsk State University Information Security and Cryptography Department – Member of TSU's SiBears Capture the Flag team 2

- 3. Prologue This is a presentation of our research devoted to new covert timing channels based on HTTP cache headers We discovered previously unknown techniques and introduced them on the ZeroNights and SibeCrypt security conferences in 2014 In the current list of «Top 10 Web Hacking Techniques of 2014» there are many valuable and significant attacks and, of course, we don't think that our work is the best. We are considering participation in 2014 Hacks as opportunity for feedback and information sharing 3
- 4. Summary We found and investigated previously unknown covert timing channels based on main HTTP cache headers We explored different properties of these covert channels (e.g., throughput, anonymity, reliability) We implemented most efficient ETag-based covert channel in Browser Exploitation Framework (BeEF) for covert communications Also we implemented ETag-based covert timing channel providing anonymity property to attackers in Google Drive environment 4
- 5. Introduction A covert channel is a path that can be used to transfer information in a way not intended by the system's designers (CWE-514) A covert storage channel transfers information through the setting of bits by one program and the reading of those bits by another (CWE-515) Covert timing channels conveys information by modulating some aspect of system behavior over time, so that the program receiving the information can observe system behavior and infer protected information (CWE-385) 5
- 6. Introduction HTTP is one of the most used protocol on the Internet so detections of the covert channels over the HTTP is an important research area HTTP timing channels have received little attention in computer security The main HTTP covert timing channel throughput is equal to 1.82 bps [1]. This channel doesn't use any HTTP mechanisms and is based on TCP/IP timing channel Server-to-Client DNS-tunnel [3] implemented in BeEF has throughput equal to 10 bit/s 6
- 7. 7 HTTP Covert Channels' Usage • Implementation of communication channels in targeted browsers (BeEF) • Botnet command and control channels • Key exchange in malicious software • Transferring of illegal content Introduction
- 8. 8 RESPONSE (SERVER) HEADERS • Last-Modified • ETag REQUEST (CLIENT) HEADERS • If-Modified-Since • If-Unmodified-Since • If-Match • If-Non-Match • If-Range General HTTP Cache Headers
- 9. 9 Covert channels can be classified as client – server channels and server – client channels Client-server covert channels are easier to implement. Server-client channels are more complicated and most of them are timing channels For example, covert storage channel via If-Range header can be implemented by the following way Directions of Covert Channels GET / HTTP/1.1 Host: evil.com If-Range: 120c7bL-32bL-4f86d4105ac62L … Hex-encoded data
- 10. Last-Modified Response Header 10 Last-Modified HTTP header stores a date of the last web entity's modification HTTP/1.1 200 OK Server: nginx/1.1.19 Date: Wed, 02 Apr 2014 14:33:39 GMT Content-Type: text/html Content-Length: 124 Last-Modified: Wed, 02 Apr 2014 14:33:39 GMT Connection: keep-alive (data) Request Response GET / HTTP/1.1 Host: evil.com
- 11. ETag Response Header 11 The ETag value is formed from the hex values by he following way HTTP/1.1 200 OK Server: Apache/2.2.22 (Ubuntu) Date: Wed, 02 Apr 2014 14:33:39 GMT

Content-Type: text/html Content-Length: 124 ETag: 120c7bL-32bL-4f86d4105ac62L Connection: keep-alive (data) Request Response GET / HTTP/1.1 Host: evil.com 120c7bL-32bL-4f86d4105ac62L file's inode size last-modified time (mtime)

- 12. Common Usage of Cache Request Headers 12 HTTP cache headers allows to web-browsers not to download a page if it hasn't been changed since the certain time Request Page has been changed HTTP/1.1 200 OK (page data) Page has not been changed HTTP/1.1 304 OK (only headers) GET / HTTP/1.1 Host: evil.com If-Modified-Since: Wed, 02 Apr 2014 14:33:39 GMT (other headers) GET / HTTP/1.1 Host: evil.com If-None-Match: 120c7bL-32bL-4f86d4105ac62L (other headers)
- 13. Common Usage of Cache Request Headers 13 Second pair of headers does the same as previous but with logically inverse condition Request Page has been changed HTTP/1.1 412 OK (page data) Page has not been changed HTTP/1.1 200 OK (only headers) GET / HTTP/1.1 Host: evil.com If-Unmodified-Since: Wed, 02 Apr 2014 14:33:39 GMT (other headers) GET / HTTP/1.1 Host: evil.com If-Match: 120c7bL-32bL-4f86d4105ac62L (other headers)
- 14. DFD Threat Model 14 read write writet Server page.html Zombie read writet read write 2 different threat models Web server is fully controlled by an attacker Payload -- read-only page.html -- write-only Web server is not controlled by an attacker write Trusted Boundaries
- 15. General Covert Channels Scheme 15 HTTP request Get new header value Received '1' If the header was changed Store header value Received '0' Wait n seconds then else
- 16. 16 RESPONSE (SERVER) HEADERS • Last-Modified • ETag REQUEST (CLIENT) HEADERS • If-Modified-Since • If-Unmodified-Since • If-Match • If-Non-Match • If-Range General HTTP Cache Headers
- 17. Last-Modified Based Channels 17 HTTP request Get Last-Modified header value Received '1' If the header value was changed Store header value Received '0' Wait n seconds then else Last-Modified header value covert channel Last-Modified: Wed, 02 Apr 2014 14:33:39 GMT
- 18. Classification 18 Covert Timing Channels based on HTTP-date entities • Based on Last-Modified header • Based on If-Modified-Since header • Based on If-Unmodified-Since header Covert Timing Channels based on ETag entities • Based on ETag header • Based on If-Match header • Based on If-None-Match header
- 19. Last-Modified based Channel 19 Zombie requests page.html and receives the HTTP response that contains initial Last-Modified value HTTP- date0 Server performs read or write access to the page.html To obtain 1 bit of information Zombie request page.html again and compares the new Last-Modified value HTTP- date1 with the old one If HTTP-date1 and HTTP-date0 is not the same, so the Server has sent 1, otherwise Server has sent 0
- 20. If-Modified-Since based Channel 20 Covert channel based If-Modified-Since header If-Modified-Since: Wed, 02 Apr 2014 14:33:39 GMT If-Modified request Received '1' If HTTP code is "200" Store header value Received '0' Wait n secondsthen else
- 21. If-Unmodified-Since based Channel 21 If-Unmodified request Received '1' If HTTP code is "412" Store header value Received '0' Wait n secondsthen else Covert channel based on If-Unmodified-Since header If-Unmodified-Since: Wed, 02 Apr 2014 14:33:39 GMT
- 22. ETag based Channel 22 Zombie requests page.html and receives the HTTP response that contains initial ETag value entity-tag0 Server performs read or write access to the page.html To obtain 1 bit of information Zombie request page.html again and compares the new ETag value

entity-tag1 If entity-tag1 and entity-tag0 is not the same, so the Server has sent 1, otherwise Server has sent 0

- 23. ETag based Channel 23 Covert channel based on ETag header ETag: 120c7bL-32bL-4f86d4105ac62L HTTP request Get ETag header value Received ‘1’ If the header value was changed Store header value Received ‘0’ Wait n seconds then else
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- 25. ETag based Channel 25 Covert channel based If-Match header If-Match: 120c7bL-32bL-4f86d4105ac62L If-Match request Received ‘1’ If HTTP code is “412” Store header value Received ‘0’ Wait n secondsthen else
- 26. Software Implementation In tons of possible ways we focused on • Python – Socket library • C++ – Boost ASIO library • C – simple C socket library We chose C due to its highest performance (among these ways) and decent stability First threat model was chosen because of its minimal requirements 26
- 27. Issues Issue Solution Server-client synchronization Special synchronizing function Different time of requests Dynamic sleep time Lateness after sleep “Active” sleep High CPU load with “active sleep” “Dynamic” and “active” sleep combination 27 Some problems we solved during implementation
- 28. 28 Send HTTP request Get host response If page has been changed then else Necessity of synchronization “read” (web client) and “write” (host) services Solution Synchronizing function that does requests at a maximum speed (without sleep) Issues
- 29. 29 Different time of requests can break services synchronization Solution Dynamic sleep time equals to sleep_time – diff_time Calculate time took for request diff_time Sleep (sleep_time – diff_time) μs Issues
- 30. 30 Inaccurate sleep - after sleep (usleep() is used) the program can awake with 10-200μs lateness Solution: Use “active sleep” - calculation time difference between last request and current moment while it is less than sleep_time Issues Calc diff_time thenelse If diff_time < sleep_time
- 31. 31 High CPU load with “active sleep” Solution Combine “active” and “dynamic” sleep Calculate diff_time If diff_time < CONST thenelse Sleep (sleep_time – CONST – request_time) where CONST is constant about 1000 μs (or less depending on PC performance) Issues
- 32. 32 Sleep time Min start sequence Avg sequence Max sequence Speed Accuracy 1 second 3200 bits 8848 bits 19712 bits 1bit/s 99,82% 2 seconds 3400 bits 10145 bits 22143 bits 0.5 bit/s 99,87% • C-based implementation in the first threat model • Min start sequence – minimum number of bits passed from the beginning of a conversation till the first mistake • Avg and Max sequence – number of bits passed without any mistakes in a row in average and at best • Accuracy – percent of correctly transmitted bits Experiment 1
- 33. 33 Sleep time Min start sequence Avg sequence Max sequence Speed Accuracy 1 second 3200 bits 8848 bits 19712 bits 1bit/s 99,82% 0.5 seconds 2400 bits 8142 bits 18123 bits 2 bit/s 99,5% • C-based implementation in the first threat model • ETag contains mtime (last modified time with microsecond accuracy), so theoretical channel capacity is bigger than its practically possible one. • Maximum practical speed of the covert channels is about 1 bit per (2L+T)

seconds, where L is HTTP latency between u2 and s1 and T is a time that is needed for auxiliary operations Experiment 2

- 34. Google Drive API Anonymity Channel 34 Most of the cloud services for file hosting like Dropbox, Google Drive and others allow users to operate with files' ETags and other cache-control headers So it is possible to implement ETag based covert timing channel in the first threat model: there are channel processes Server(attacker1) and Zombie (attacker2) on different hosts and fully trusted web server https://drive.google.com/drive/ with some file hosted on it. The only requirement for that is file should be accessible for writing by attacker1 and for reading by attacker2
- 35. Google Drive API Anonymity Channel Covert channel's logic is the same as before: • attacker1 sends a request to Google Drive API POST https://www.googleapis.com/drive/v2/files/fileId/touch to modify file's last access time (and hence ETag) • attacker2 sends a request to Google Drive API GET https://www.googleapis.com/drive/v2/files/fileId to get file's metadata (including ETag) This channel has property that provides anonymity for communications between Server and Zombie 35
- 36. Experiment 3 36 Message length 256 bit 512 bit 1024 bit 2048 bit 4096 bit Accuracy 99.87% 99.84% 99.8% 99.8% 99.78% Average throughput 2.92 bit/s 2.9 bit/s 2.88 bit/s 2.88 bit/s 2.86 bit/s Google Drive API anonymity covert channel based on ETag header
- 37. Advantages in the First Threat Model 37 • Anonymity • Does not modify common HTTP request structure • Does not require web-server modifications • Any read-only activity on web page that is used by the channel do not break its work • Information flow looks like something refreshes a web page every n seconds • Covert channels based on If-* headers can work even if Last-Modified or Etag are disabled
- 38. Second Threat Model In the second threat model we can avoid necessity of client-server synchronization by waiting for the request and responding directly 38 Send new header value Send old header value If current message bit is '1' Store header value then else WAIT for HTTP request
- 39. Experiment 4 C-based client, Apache + PHP-based server 39 Header Network Average HTTP ping Speed ETag Local host 0.55 ms 986 bit/s «Digital Ocean» DC LAN 1.63 ms 845.65 bit/s LAN 6.9 ms 295.69 bit/s Internet 113.2 ms 13.09 bit/s
- 40. Experiment 5 C-based client, Flask + Python-based server 40 Header Network Average HTTP ping Speed ETag Local host 0.55 ms 981 bit/s «Digital Ocean» DC LAN 1.63 ms 865.83 bit/s LAN 6.9 ms 293.9 bit/s Internet 103.2 ms 14.39 bit/s
- 41. Advantages in Second Threat Model 41 • Does not modify common HTTP request structure • Information flow looks like something refreshes a web page every n seconds • Higher throughput • Reliability • Simplicity • This approach is applicable for implementation of covert channels based on HTTP cache headers in browsers
- 42. Covert Channels in Browsers Issues • Lack of any "sleep" function • Low accuracy of existing time management functions • Difficulties with synchronization of covert channel's server and client So implementation of the used model is pointless, but it is possible to implement covert channels in these restrictions using controlled web server in the second threat model 42

- 43. Implementation of ETag-based covert channel in browser (client on JavaScript) 43 Experiment 6 Header Server Average HTTP ping Throughput Last-Modified 0.045 ms 70 ms 1 bit/s Last-Modified 18 ms 68 ms 1 bit/s ETag Python 66 ms 11.51 bit/s ETag PHP 72 ms 10.8 bit/s
- 44. Covert Channels in BeEF “BeEF allows the professional penetration tester to assess the actual security posture of a target environment by using client-side attack vectors.” The main idea was proposed in Kenton Born’s paper “Browser-based covert data exfiltration” [2] and is being used in BeEF [3] To investigate covert timing channels in browsers we implemented server-to-client DNS and ETag Tunnels using AJAX and then added them to BeEF 44
- 45. ETag-based timing channel in BeEF 45 Issue Solution Server-client synchronization Client does special request to begin conversation End of message determination Client receive some special HTTP code in response, e.g. 404 – Not Found or 403 - Forbidden Single client communication only Open a session that stores transferring bit number for each client
- 46. ETag-based timing channel in BeEF 46 ETag Tunnel in BeEF consists s of classic two parts • extension on Ruby, that implements server side logic via couple of web pages mounted to BeEF webserver • module on JavaScript, that is responsible for receiving information from C&C BeEF server at zombie Sources • https://github.com/beefproject/beef/tree/master/modules/i pec/etag_client • https://github.com/beefproject/beef/tree/master/extension s/etag
- 47. Implementation of ETag-based covert channel in browser (client on JavaScript) 47 Experiment 7 Network Average ping Average HTTP ping 256 bit 1024 bit Local host 0.045 ms 0.6 ms 10.11 bit/s 9.9 bit/s Local network 18 ms 19.8 ms 10.3 bit/s 9.78 bit/s Internet 176 ms 360.9 ms 5.09 bit/s 4.97 bit/s
- 48. 48 Proof of Concept http://youtu.be/W2qWA7XUzGQ https://github.com/beefproject/beef
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